

# Prepared reports. Faster review. Better operational context.

A SharePoint-first manufacturing intelligence platform for supervisor and manager decision support.

## How people use the outputs

**Shift Drafts** automate collection, reconciliation, organization, and spreadsheet preparation. Supervisors review exceptions, add context, and finalize faster.

**Daily Production Reports** help managers understand production, molding, labor, quality, plan attainment, and downtime; investigate constraints; prioritize follow-up; and inform staffing, plan, maintenance, or coordination changes.

**Weekly Production and Downtime Reports** show trends, comparisons, and recurring problems for planning, improvement, meetings, reviews, and handoffs.

| INPUT                                    | PROCESS                                   | OUTPUT                                            | REVIEW + ACTION                     |
|------------------------------------------|-------------------------------------------|---------------------------------------------------|-------------------------------------|
| Forms, SharePoint, files, machine events | Ingest, reconcile, link sources, validate | Shift drafts, daily and weekly reports, dashboard | People interpret context and decide |

## Operational intelligence and value

**Chatbot:** source-grounded retrieval with validated citations and deterministic fallback, estimated to reduce representative retrieval time about 90%. **Report doctor:** read-only health and exception diagnostics. **Governed oracle:** contract-bound interpretation and validation.

**Modeled labor opportunity:** 20 supervisor plus 10 manager hours per week, about \$52K annual labor value. These are planning signals—not booked savings or autonomous decisions.

# Architecture and reliability evidence

## SharePoint and Forms intake

Five Power Automate flows cover reminders, machine events, machine whiteboard submissions, molding-report archive, and labor notes. SharePoint lists and libraries retain operational inputs and outputs.

## Source-linked reporting core

Python ingestion and reconciliation normalize evidence into eight PostgreSQL serving tables. Workbooks and the API/SPFx dashboard expose prepared reporting views with source lineage.

## Contracts and state

Report identity, expected-output state, material identity, source lineage, semantic certification, publication state, and readback settlement remain explicit. Missing values remain distinct from zeros; ambiguous evidence cannot silently advance state.

## Fail-closed delivery and recovery

Artifact validation and GET readback precede settlement. Bounded repair operates only on eligible evidence; one-writer releases prevent overlap; exact rollback restores the last known-good immutable revision.

# Operational intelligence

## Source-grounded chatbot

Retrieval is bounded to governed report evidence. Responses are structured, citations are validated, and deterministic fallback refuses unsupported answers. The documented 20–30 minute to 1–2 minute scenario represents an estimated 90% retrieval improvement.

## Read-only report doctor

Expected-output health, missing or late reports, exception classifications, and settlement/readback state are observable without granting mutation, repair, or publication authority.

## Governed oracle

Interpretation is evaluated against contracts, lineage, identity, state, and semantic boundaries. Missing, stale, or ambiguous evidence fails closed. Human operators retain decision authority.

## Impact methodology

**Labor:** 30 hours/week × 52 weeks using documented supervisor/manager rate assumptions, about \$52K annual labor value. **Retrieval:** 20–30 minutes reduced to 1–2 minutes, about 90%. **Downtime upper bound:** 5 minutes × 1,000 widgets/hour/machine × 52.5 events/week × 52 weeks = up to 227,500 widgets; at \$0.90/widget, approximately \$205K contribution-margin opportunity.

All are planning scenarios—not booked savings, realized production, guaranteed margin, or autonomous outcomes.

## Public/private boundary

Public assets use synthetic data and sanitized architecture. Tenant identifiers, employee data, production URLs, credentials, and exact operating details remain in the private handoff.